A Systematic Benchmark of Supervised and Unsupervised Graph Embedding Methods

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***Abstract*—Graph embedding methods provide low dimensional representations of graphs for downstream learning tasks. In this work, we compare unsupervised and supervised graph embedding techniques on real world benchmark datasets. We evaluate Graph2Vec, NetLSD and the Graph Isomorphism Network (GIN) on MUTAG, ENZYMES and IMDB-MULTI using graph classification and clustering, considering accuracy, computational efficiency and scalability. Results show that GIN achieves the highest classification accuracy at a higher computational cost, while Graph2Vec offers a favorable balance between performance and efficiency. NetLSD demonstrates competitive clustering performance with minimal overhead. Overall, each method shows distinct trade offs and maintains reasonable robustness under structural perturbations providing practical guidance for method selection under different data and computational constraints.**

***Index Terms*—Graph2Vec, NetLSD, GIN, Graph Embedding, Graph Classification, Graph Clustering**

I. Introduction

Graphs are widely used to model relationships and interactions in data from multiple domains, such as network analysis, bioinformatics, and chemistry. Many learning tasks are defined at the graph level. However, graphs exhibit irregular structures with variable sizes and features, which pose challenges for traditional machine learning algorithms.

Graph embedding methods address these limitations by mapping entire graphs into fixed-size vector representations, enabling standard machine learning techniques to be applied efficiently to downstream tasks. Early approaches based on handcrafted features or graph kernels suffer from high computational complexity or limited expressiveness [6], [7]. More recent methods automatically learn graph representations and can be broadly categorized into unsupervised [1], [2] and supervised techniques [3].

Despite their success, supervised methods require labeled data and typically incur higher computational costs due to iterative training, whereas unsupervised methods do not require labels and are generally more efficient. However, systematic comparisons between these two paradigms remain limited, especially on small benchmark datasets where robustness and efficiency are critical [9], [10].

In this work, we focus on the EMB1 setting and compare unsupervised and supervised graph embedding methods on small benchmark datasets from TUDataset. We evaluate Graph2Vec, NetLSD and GIN under a unified experimental framework, considering not only classification performance but also clustering quality, computational cost and stability to structural perturbations.

Our contributions are summarized as follows:

- A unified and fair experimental framework for comparing supervised and unsupervised graph embedding methods on small datasets
- An analysis of performance–efficiency trade offs, and
- An evaluation of embedding robustness under structural perturbations.

To guide our empirical evaluation, we address the following research questions:

- **RQ1:** How do unsupervised graph embedding methods compare to supervised GNN-based approaches in terms of classification performance on small benchmark datasets?
- **RQ2:** How well do different graph embeddings capture graph similarity, as reflected by clustering quality?
- **RQ3:** What are the computational efficiency trade-offs between unsupervised and supervised methods?
- **RQ4:** How robust are graph embeddings to structural perturbations of the input graphs?

II. Graph Embedding Methods

This section describes the graph embedding methods evaluated in this work.

Algorithm 1 Graph2Vec

Require: Graph set $\mathcal{G}$, WL depth D, dim δ, epochs e, lr α
Ensure: Graph embeddings $\Phi \in \mathbb{R}^{|\mathcal{G}| \times \delta}$
Initialize $\Phi \sim \mathcal{U}(-0.5/\delta, 0.5/\delta)$
for *epoch* $= 1$ to e **do**
  Shuffle $\mathcal{G}$
  for each $G_i \in \mathcal{G}$ **do**
    for $d = 0$ to D **do**
      for each node $n \in G_i$ **do**
        $sg \leftarrow \text{WLSubgraph}(n, G_i, d)$
        Update Φ_i via Skip-Gram(sg, α)
      end for
    end for
  end for
end for
return Φ

Algorithm 2 WLSubgraph

Require: Node n, graph $G = (N, E, \lambda)$, depth d
Ensure: Rooted subgraph label sg
$sg \leftarrow \lambda(n)$
for $i = 1$ to d **do**
  $sg \leftarrow \text{hash}\big(sg \,\|\, \{\lambda(u) : u \in \mathcal{N}(n)\}\big)$
end for
return sg

Algorithm 3 NetLSD (Heat Trace Signature)

Require: Graph $G = (V, E)$, times $T = \{t_k\}_{k=1}^{K}$, Laplacian type $\mathcal{L}$, eigenvalues m
Ensure: Graph signature $s \in \mathbb{R}^K$
Build adjacency A and degree D
$L \leftarrow \begin{cases} I - D^{-1/2} A D^{-1/2}, & \mathcal{L} = \text{norm} \\ D - A, & \mathcal{L} = \text{comb} \end{cases}$
Compute smallest m eigenvalues $\{\lambda_i\}_{i=1}^{m}$ of L
for $k = 1$ to K **do**
  $s_k \leftarrow \sum_{i=1}^{m} e^{-t_k \lambda_i}$
end for
$s \leftarrow \log(s)$ ▷ optional
$s \leftarrow \text{Normalize}(s)$ ▷ optional
return s

Algorithm 4 GIN Forward Pass

Require: Graph $G = (V, E)$, node features $\{h_v^{(0)}\}$, layers L
Ensure: Graph embedding z_G
for $k = 1$ to L **do**
  for all $v \in V$ **do**
    $h_v^{(k)} \leftarrow \text{MLP}^{(k)}\Big((1 + \epsilon^{(k)}) h_v^{(k-1)} + \sum_{u \in \mathcal{N}(v)} h_u^{(k-1)}\Big)$
  end for
end for
$z_G \leftarrow \text{READOUT}(\{h_v^{(L)}\})$
return z_G

A. Graph2Vec

We begin with Graph2Vec [1], an unsupervised whole-graph embedding method that learns fixed-length vector representations in a manner analogous to Doc2Vec for documents. The core idea is to treat each graph as a document and its rooted subgraphs, corresponding to local node neighborhoods, as words. Under this formulation, graphs that share similar structural patterns are expected to be mapped to nearby points in the embedding space.

More specifically, Graph2Vec first applies the Weisfeiler–Lehman (WL) relabeling procedure to extract rooted subgraphs around each node up to a specified depth D (Algorithm 2). These rooted subgraphs are then aggregated across all graphs and WL iterations to form a global vocabulary, referred to as a *bag of subgraphs*. Finally, a skip-gram model is trained to learn graph-level embeddings by maximizing the likelihood of observing the rooted subgraphs associated with each graph (Algorithm 1).

B. NetLSD

NetLSD [2] represents a graph using the spectrum of its graph Laplacian, capturing global structural information. The underlying idea is that the eigenvalues of the Laplacian matrix encode important properties of the graph structure. Rather than using the raw eigenvalues directly, NetLSD summarizes the spectral information through a diffusion process, specifically via the heat kernel (Algorithm 3).

A key advantage of NetLSD is its permutation invariance and scalability. To improve computational efficiency, the method approximates the Laplacian spectral distribution using stochastic trace estimation, thereby avoiding full eigendecomposition. As a result, NetLSD maintains stable performance even for large graphs.

C. GIN

The Graph Isomorphism Network (GIN) [3] is a supervised graph neural network designed to maximize the discriminative power of GNNs. In GIN, node representations are iteratively updated by aggregating information from neighboring nodes. At each layer, node features are updated using a sum aggregation function followed by a multilayer perceptron (MLP), which enables the model to distinguish different graph structures effectively (Algorithm 4). The resulting node embeddings are then aggregated via a global pooling operation to obtain a graph-level representation. This representation is subsequently passed to a classifier and trained end-to-end using labeled data (Algorithm 5).

In contrast to unsupervised methods, GIN learns task-specific embeddings that are optimized directly for classification performance. While this typically leads to higher accuracy, it requires labeled datasets, which are often difficult to obtain, and incurs higher computational costs due to iterative training.

III. Datasets

The experiments in this work are conducted on three datasets from TUDataset: MUTAG, ENZYMES, and IMDB-MULTI. These datasets are small but diverse, covering domains such as chemistry, biology, and social networks. They are standard benchmarks in the graph representation learning literature [1]–[3] and are commonly used to evaluate graph embedding and classification methods.

Algorithm 5 Training GIN

Require: Dataset $\mathcal{D}$, epochs E, lr α, batch size B
Ensure: Trained parameters Θ
Initialize Θ
for *epoch* $= 1$ to E **do**
  for mini-batch $\mathcal{B} \subset \mathcal{D}$ **do**
    for all $(G, y) \in \mathcal{B}$ **do**
      $z_G \leftarrow \text{GINForward}(G)$
      $\hat{y} \leftarrow \text{Classifier}(z_G)$
    end for
    $\mathcal{L} \leftarrow \frac{1}{|\mathcal{B}|} \sum \ell(\hat{y}, y)$
    Update Θ (e.g., Adam)
  end for
end for
return Θ

A. MUTAG

MUTAG [4] is a molecular graph dataset consisting of chemical compounds, where each graph represents a molecule, nodes correspond to atoms, and edges represent chemical bonds. The task is to classify molecules into two classes according to their mutagenic effect on a bacterium. The dataset contains a relatively small number of graphs with discrete node labels representing atom types. Graphs in MUTAG are typically small and sparse, making the dataset suitable for fast experimentation.

B. ENZYMES

The ENZYMES [5] dataset is derived from protein tertiary structures and consists of graphs representing enzymes. Nodes correspond to secondary structure elements, while edges indicate spatial proximity or interactions between residues. Each graph is labeled according to the enzyme's functional class, resulting in a multiclass classification task with six classes.

C. IMDB-MULTI

IMDB-MULTI [5] is a social network dataset constructed from movie collaboration data. Each graph represents the ego network of actors appearing in a movie, where nodes correspond to actors and edges represent co-appearance relationships. Unlike molecular and biological datasets, IMDB-MULTI does not provide node labels or attributes.

IV. Experimental Setup

A. Experimental Pipeline

Our experimental pipeline follows a unified and reproducible procedure across all datasets and methods. For each dataset, graphs are first converted into fixed-length vector representations using the selected graph embedding methods. These embeddings are then used for downstream tasks including graph classification, clustering, and stability analysis.

For unsupervised methods (Graph2Vec and NetLSD), graph embeddings are computed independently of any labels and stored for subsequent evaluation. Classification is performed by training standard machine learning models on top of the learned graph vectors. For the supervised method (GIN), graph embeddings are learned jointly with the classification objective through end-to-end training.

All experiments are conducted consistently across embedding dimensions of 64, 128, and 256 in order to study the effect of representation capacity on performance and computational cost.

B. Data Splits and Reproducibility

For each dataset and embedding dimensionality, embeddings are split into 80% training and 20% test sets. To ensure fair comparison across methods, the same data splits are used for all embedding approaches within each experimental configuration.

Random seeds are fixed across all experiments to reduce variability due to stochastic effects. All reported results are obtained using identical splits, configurations, and evaluation scripts for each method.

C. Implementation and Software

Unsupervised graph embeddings are generated using publicly available implementations of Graph2Vec and NetLSD. The resulting embeddings are stored in CSV format, with one row per graph containing the graph label and the corresponding embedding vector.

Downstream classification for unsupervised embeddings is performed using `scikit-learn`. Prior to training, all embeddings are standardized using `StandardScaler`. The following classifiers are employed: Support Vector Machines with RBF kernel, Multilayer Perceptrons, Random Forests, Logistic Regression, Gradient Boosting, and a Voting Classifier combining heterogeneous models.

The supervised GIN model is implemented using PyTorch Geometric and follows the architecture and training protocol described in Algorithms 4 and 5. GIN models consist of stacked GINConv layers parameterized by multilayer perceptrons, followed by batch normalization, ReLU activation, dropout, and residual connections. Graph-level representations are obtained using additive global pooling. Jumping Knowledge with concatenation is employed to aggregate information from multiple layers.

Several training options are enabled to improve stability and generalization, including feature normalization, graph-level feature augmentation, and label smoothing. Early stopping is applied during training to prevent overfitting on small datasets.

D. Hyperparameter Configuration

For Graph2Vec and NetLSD, experiments are conducted using embedding dimensions of 64, 128, and 256. Classifier hyperparameters are adapted to dataset size to mitigate overfitting on small benchmarks.

For GIN, multiple training configurations are explored to analyze the effect of model capacity and regularization. Embedding dimensions are set to 64, 128, and 256. Training is performed using learning rates of 1×10^{-3}

and 5×10^{-4}. Dropout rates of 0.0 and 0.4 are evaluated to assess the impact of regularization on small datasets. The number of GIN layers is selected per dataset, with shallower architectures used for social network graphs and deeper architectures for molecular and biological graphs. Specifically, the number of layers is set to 3 for IMDB-MULTI, 5 for MUTAG, and 6 for ENZYMES. Batch sizes are set to 64 for MUTAG and IMDB-MULTI, and 32 for ENZYMES.

In addition to architectural variations, several training options and design choices are enabled to improve stability and generalization. Graph-level feature augmentation is applied by concatenating simple structural statistics, including the number of nodes, number of edges, and graph density, to the learned graph embeddings. Additive global pooling is employed to aggregate node representations into graph-level embeddings, as it has been shown to provide stable performance across datasets. Feature normalization is applied after node feature construction to improve optimization behavior. Furthermore, label smoothing with a factor of 0.05 is applied to the cross-entropy loss in order to reduce overconfidence during training.

Training is conducted for up to 200, 300, 500, or 800 epochs depending on the dataset, with early stopping enabled to mitigate overfitting and reduce sensitivity to the maximum number of epochs.

This hyperparameter exploration allows us to analyze how architectural depth, embedding dimensionality and regularization affect both classification accuracy and the quality of the learned graph embeddings.

E. Hardware Configuration

All experiments are executed on a Linux-based system equipped with a multi-core CPU and sufficient memory to support large-scale embedding computation and training. Supervised GIN models are trained using GPU acceleration when available. Memory usage is monitored during execution to capture peak allocation and resident set size for computational efficiency analysis.

V. Evaluation Methodology

A. Graph Classification Evaluation

We evaluate graph embeddings on three benchmark datasets from TUDataset. Unsupervised methods (Graph2Vec and NetLSD) are evaluated under a common downstream classification pipeline, while the supervised GIN model follows its native end-to-end training protocol.

Classification performance is assessed using Accuracy, F1-score, and Area Under the ROC Curve (AUC). In addition to predictive performance, we report computational metrics including training time, embedding generation time, and memory consumption, measured via peak memory usage and resident set size (RSS). To study the effect of representation capacity, all methods are evaluated using embedding dimensions of 64, 128, and 256. Smaller embedding dimensions were initially considered to reduce computational cost and memory consumption. However, evaluating higher-dimensional representations allows us to examine the trade-off between efficiency and representational capacity, as discussed in the results section.

B. Evaluation of Unsupervised Embeddings

For unsupervised methods, graph embeddings are evaluated by training standard classifiers on top of the learned representations. Embeddings are split into 80% training and 20% test sets for each dataset and embedding dimensionality.

Prior to classification, embeddings are standardized using a z-score normalization (sklearn's StandardScaler) to ensure consistent feature scaling, which is particularly important for distance- and margin-based classifiers. The following classifiers are used for evaluation:

- Support Vector Machine (SVM) with an RBF kernel and probability calibration enabled for AUC computation,
- Multilayer Perceptron (MLP), with hidden layer sizes adapted to the embedding dimensionality, using deeper architectures for higher-dimensional embeddings,
- Random Forest classifier, with the number of trees and maximum depth adjusted according to dataset size to mitigate overfitting on small benchmarks,
- Logistic Regression with a multinomial formulation for multiclass datasets, and
- Gradient Boosting classifier with moderate tree depth.

In addition, an ensemble Voting Classifier combining Random Forest, Gradient Boosting and SVM is employed to assess the benefits of combining heterogeneous decision boundaries.

C. Evaluation of Supervised Graph Embeddings

For supervised evaluation, GIN produces graph-level representations through global pooling of node embeddings and is trained end-to-end using labeled data. The resulting graph representations are evaluated directly through the model's classification output, without training additional downstream classifiers.

D. Clustering Evaluation

To assess how well graph embeddings preserve graph-to-graph similarity, we evaluate clustering performance on the learned representations. Clustering quality reflects the ability of an embedding to capture global structural properties in the absence of label supervision. Embeddings that encode global structural properties are expected to perform better at clustering than those optimized for discriminative classification.

E. Stability Analysis

Real-world graph data are often noisy, with missing edges or corrupted node attributes. To assess the robustness of graph embeddings under such perturbations, we perform a stability analysis by introducing controlled modifications to graph structures and node features.

For each dataset, perturbed versions of the graphs are generated using two types of perturbations. First, edge perturbation is applied by randomly removing 10% of existing edges and adding an equal number of randomly generated edges. Second, node attribute perturbation is introduced by randomly shuffling node features between pairs of nodes within each graph.

After perturbation, graph embeddings are recomputed using Graph2Vec, NetLSD, and GIN with the same scripts and configurations as in the original experiments. This ensures that any observed differences in the embeddings are attributable solely to the introduced perturbations.

Embedding stability is quantified using two complementary similarity measures. Cosine similarity is used to assess directional consistency between the original and perturbed embeddings,

$$\cos(z_i, z_i') = \frac{z_i^\top z_i'}{\|z_i\|_2 \|z_i'\|_2},$$

while the Euclidean distance measures absolute deviation,

$$\|z_i - z_i'\|_2.$$

For each dataset, method, and experimental configuration, we report the mean, median, and standard deviation of both cosine similarity and Euclidean distance across all graphs. To evaluate the practical impact of perturbations, we additionally repeat the graph classification and clustering experiments on the perturbed datasets using the same evaluation protocols.

VI. Results

A. Before Permutation

For the graph classification tasks before any perturbations, we present the performance of each embedding method across the three TUDatasets with accuracy, F1-score and AUC metrics for each method in Tables I, II and III. We preferered to show only the best test configuration per dataset x dim results and for the epochs 200 and 300 (for GIN) for the reason we mentioned in earlier section.

For the MUTAG, we can see that GIN excels in classification accuracy, but NetLSD provides superior probabilistic separation as indicated by its higher AUC scores. AUC slightly decreases with increasing dimensions for GIN, showing that higher classification accuray does not always mean that the method has confidence in its predictions. So, our conclusion is that spectral info captured by NetLSD is beneficial for this dataset. For ENZYMES, the gap between GIN and the other two methods is clearer in all metrics. GIN clearly dominates here and AUC metric confirms that GIN is more confident in its predictions. For IMDB-MULTI, when node features are absent, GIN and NetLSD have similar behavior, with GIN slightly ahead.

TABLE I: GIN classification performance on EMB1 datasets (epochs $\in \{200, 300\}$). Best configuration per embedding dimension is reported.

Dataset	Dim	Epochs	Acc	F1	AUC
MUTAG	64	200	0.842	0.825	0.885
	128	200	0.842	0.808	0.885
	256	200	0.895	0.864	0.859
ENZYMES	64	300	0.433	0.428	0.703
	128	300	0.483	0.492	0.744
	256	300	0.617	0.615	0.809
IMDB-MULTI	64	200	0.500	0.496	0.686
	128	200	0.500	0.486	0.678
	256	200	0.507	0.478	0.714

TABLE II: Best Graph2Vec classification results per dataset and embedding dimension. For each dataset–dimension pair, the classifier with the highest test accuracy is reported.

Dataset	Dim	Model	Acc	F1	AUC
MUTAG	64	RF	0.711	0.656	0.775
	128	RF	0.763	0.754	0.797
	256	RF	0.737	0.731	0.745
ENZYMES	64	Ens	0.258	0.252	0.571
	128	RF	0.225	0.219	0.600
	256	RF	0.317	0.310	0.591
IMDB-MULTI	64	RF	0.437	0.435	0.601
	128	GB	0.427	0.426	0.583
	256	SVM	0.433	0.426	0.599

So overall, GIN is the most effective method and NetLSD is second best, while Graph2Vec shows the weakest performance across all datasets and metrics. The results confirm that supervised GNNs provide the highest predictive power, while spectral descriptors such as NetLSD remain surprisingly strong for unsupervised graph representation learning.

The experimental results reveal clear trade offs between classification accuracy and computational efficiency across GIN, Graph2Vec and NetLSD, as illustrated in Figures 1-3. All figures are computed with the average results across the three dimensions and three datasets. We chose these 3 images (from the 39 we generated for the classification section) to summarize and make our conclusions.

Figure 1 presents a heatmap of average accuracy across methods and embedding dimensions. We can confirm that GIN consistently outperforms the other two methods across all dimensions.

The Pareto frontier in Figure 2 highlights the balance between efficiency and performance. NetLSD lies on the left most side of the optimal region, meaning that it achieves reasonable accuracy with minimal compute cost. GIN dominates with its accuracy, but with a significantlt higher compute cost. Graph2Vec is at the lower part of the graph, showing both lower accuracy from both other methods and higher compute cost, than NetLSD. From the same figure, we can see that GIN benefits monotonically from increased embedding dimensionality, achieving the

TABLE III: Best NetLSD classification results per dataset and embedding dimension. For each dataset–dimension pair, the classifier with the highest test accuracy is reported.

Dataset	Dim	Model	Acc	F1	AUC
MUTAG	64	Ens	0.895	0.892	0.908
	128	Ens	0.868	0.867	0.883
	256	LR	0.868	0.867	0.929
ENZYMES	64	RF	0.300	0.302	0.666
	128	RF	0.308	0.312	0.653
	256	RF	0.325	0.324	0.673
IMDB-MULTI	64	SVM	0.480	0.477	0.625
	128	RF	0.463	0.465	0.633
	256	RF	0.457	0.456	0.648

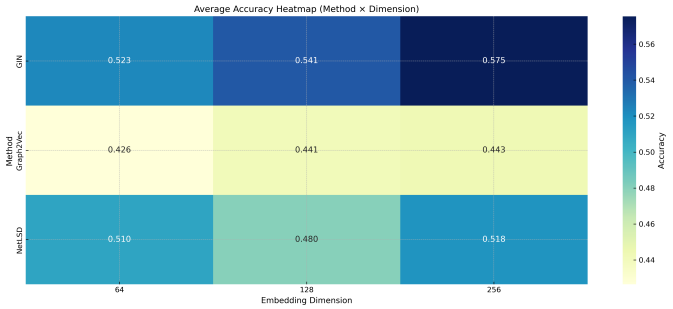


Fig. 1: Average accuracy heatmap across methods and embedding dimensions.

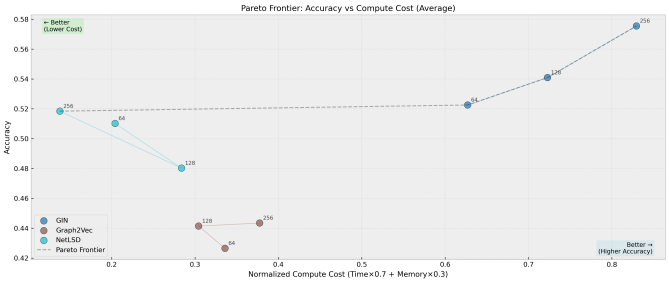


Fig. 2: Pareto frontier illustrating the trade-off between accuracy and normalized compute cost.

TABLE IV: Best Graph2Vec clustering per dataset and dimension (highest ACC; ties by Silhouette).

Data	Dim	Alg	NMI	ARI	ACC	Sil
MUTAG	64	spec	0.048	-0.019	0.532	0.285
	128	km	0.033	-0.019	0.521	0.305
	256	spec	0.037	-0.027	0.511	0.315
ENZ	64	km	0.027	0.006	0.223	0.205
	128	km	0.030	0.010	0.220	0.283
	256	spec	0.021	0.005	0.220	0.210
IMDB	64	km	0.013	0.011	0.402	0.506
	128	km	0.011	0.010	0.399	0.552
	256	km	0.011	0.010	0.400	0.584

highest average accuracy at 256 dimensions. In contrast, Graph2Vec and NetLSD reach their peak accuracy at 128 dimensions, beyond which performance declines, indicating diminishing returns with higher dimensions.

Figure 3 breaks down the compute cost into embedding time, training time and peak memory usage. Training time (Figure 3b) is dominated by GIN and increases substantially with embedding dimension. Embedding time (Figure 3a) is highest for NetLSD at lower dimensions due to its spectral computations but decreases at higher dim sizes. Peak memory usage (Figure 3c) is highest for Graph2Vec, while NetLSD remains more memory efficient.

Tables IV, V and VI are a compact version from the original clustering results tables. Best metrics are chosen based on highest ACC and silhouette as a tiebreaker.

Graph2Vec shows the weakest clustering performance. Something we expected based on its classification results and its confidence in seperations (as seen from AUC scores). Wwith the low ACC scores and zero or negative ARI scores, we can say that Graph2Vec has poor aligment with the true class labels.

NetLSD has the strongest clustering performance among the three methods. For MUTAG, it achieves high ACC and solid NMI and ARI scores indicating good alignment with true labels. For ENZYMES and IMDB-MULTI, it consistently outperforms Graph2Vec, though the overall clustering quality is moderate.

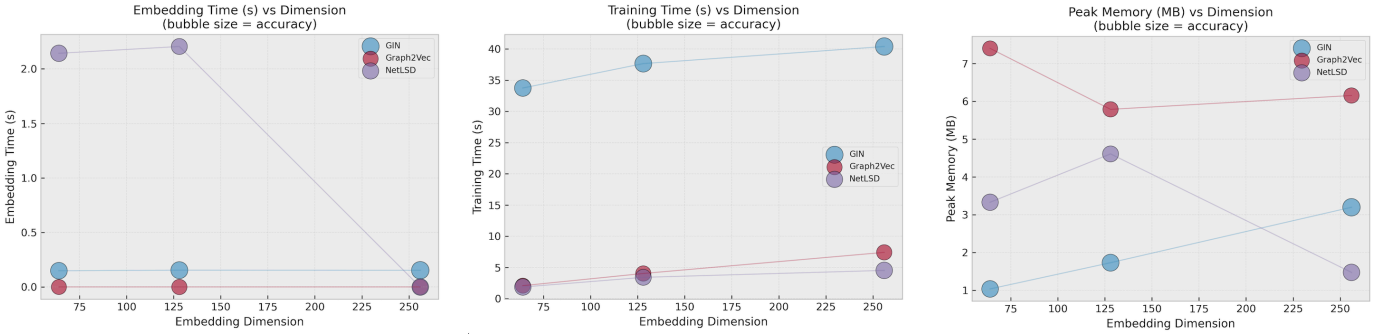
GIN creates mixed feelings here. The fact that is a supervised representation makes it not optimal for clustering tasks, meaning for unsupervised tasks. GIN performs better than Graph2Vec, but worse than NetLSD.

The experimental results for clustering evaluation are well supported by the theoretical facts. More specifically, the graph2vec method emphasizes on local rooted substructures treating graphs as unorderdered colections of neighborhood patterns. This method is effective for capturing local similarities while it lacks a mechanism for representing global structures. On the contrary NetLSD embeds grpahs using spectral summaries of the Laplacian, which are permutation invariant and capture global diffusion behaviour. Such characteristics are well suited for clustering. This exlplains why NetLSD consitently forms semantically coherent clusters despite being label-free. GIN embeddings are designed to maximise separablity rather than maintaining inter-graph distances. Consequently embeddings using GIN perform poorly during clustering. The theoretic differences are clearly represented below: »»»> ?????

B. After Permutation

So after the permutations we did where we explain in detail in the section *V*, we present the stability results for each method in Tables VII, VIII IX and X.

The stability behavior of GIN is strongly influenced by training configuration and dataset characteristics. When we optimize for max cosine (closer to value 1), we take models with dropout 0.0 and we have high cosine similarity values (0.66-0.91 range) with gradually decreasing cosine similarity as dimensionality increases. However, the corresponding L2 distances are relatively large (ranging from



(a) Embedding time (b) Training time (c) Peak memory

Fig. 3: Compute cost components as a function of embedding dimension. Bubble size indicates average accuracy.

TABLE V: Best NetLSD clustering per dataset and dimension (highest ACC).

Data	Dim	Alg	NMI	ARI	ACC	Sil
MUTAG	64	km	0.247	0.302	0.777	0.576
	128	km	0.247	0.302	0.777	0.576
	256	km	0.247	0.302	0.777	0.576
ENZ	64	spec	0.038	0.014	0.250	0.168
	128	spec	0.038	0.014	0.250	0.168
	256	spec	0.038	0.014	0.250	0.168
IMDB	64	km	0.021	0.016	0.410	0.546
	128	km	0.021	0.016	0.410	0.546
	256	km	0.021	0.016	0.410	0.546

TABLE VI: Best GIN clustering per dataset and dimension (highest ACC).

Data	Dim	Alg	NMI	ARI	ACC	Sil
MUTAG	64	km	0.244	0.236	0.745	0.445
	128	km	0.209	0.176	0.713	0.406
	256	km	0.298	0.135	0.691	0.621
ENZ	64	spec	0.082	0.028	0.282	0.029
	128	spec	0.088	0.014	0.288	0.050
	256	spec	0.221	0.128	0.365	0.076
IMDB	64	spec	0.014	0.008	0.378	-0.535
	128	spec	0.011	0.004	0.379	-0.308
	256	spec	0.010	0.006	0.391	-0.245

100 to 1800), meaning significant magnitude changes in embeddings despite relatively preserved directions.

When we optimize for min L2 (closer to value 0), we take models with dropout 0.4 and we have lower cosine similarity values (0.12-0.77 range), indicating more directional changes. However, the L2 distances are much smaller (ranging from 66 to 483), showing reduced magnitude changes. This inverse relationship between cosine similarity and L2 distance across configurations highlights the trade off between preserving direction versus magnitude in GIN embeddings under perturbations.

Graph2Vec shows strong stability across all datasets and dimensions. Cosine similarity has closer to 1 values and l2 distances are very small. This indicates that Graph2Vec embeddings are robust to structural and feature perturbations, maintaining both direction and magnitude.

TABLE VII: GIN embedding stability (best per dataset–dim by highest cosine mean, dropout = 0.0)

Data	D	LR	Cos	σ	L2	σ
MUTAG	64	1e-3	0.918	0.027	256	131
	128	1e-3	0.862	0.032	454	197
	256	1e-3	0.869	0.032	783	381
ENZ	64	5e-4	0.817	0.041	1312	1965
	128	5e-4	0.783	0.040	1100	813
	256	5e-4	0.796	0.038	1804	1743
IMDB	64	5e-4	0.664	0.070	117	160
	128	5e-4	0.671	0.072	138	259
	256	1e-3	0.666	0.066	182	363

TABLE VIII: GIN embedding stability (best per dataset–dim by lowest L2 mean, dropout = 0.4).

Data	D	LR	L2	σ	Cos	σ
MUTAG	64	5e-4	168	43	0.775	0.016
	128	1e-3	263	77	0.781	0.030
	256	5e-4	455	167	0.752	0.028
ENZ	64	5e-4	402	457	0.563	0.056
	128	1e-3	483	1835	0.297	0.085
	256	1e-3	460	1579	0.126	0.080
IMDB	64	1e-3	69	118	0.663	0.066
	128	1e-3	105	182	0.625	0.068
	256	1e-3	66	132	0.126	0.072

Finally, NetLSD shows mixed stability results. We observe that with dim sizes 256, we have high cosine similarity values and in general small L2 distances (if we compare with GIN). However, at lower dimensions (64 and 128), cosine similarity drops significantly (except for ENZYMES), even becoming negative for MUTAG. L2 distances increase with dimensionality but remain moderate relative to GIN.

So, across the three methods, Graph2Vec is the king of stability, NetLSD achieves very strong stability only at higher dim sizes and GIN shows the most sensitivity to perturbations.

Executing again the classification task after perturbing the graphs ....

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TABLE IX: Graph2Vec embedding stability under perturbations.

Data	**D**	**Cos**	σ	**L2**	σ
	64	0.899	0.018	0.133	0.052
MUTAG	128	0.910	0.023	0.095	0.039
	256	0.899	0.027	0.068	0.027
	64	0.825	0.029	0.603	0.138
ENZ	128	0.863	0.022	0.545	0.124
	256	0.896	0.020	0.487	0.117
	64	0.719	0.062	0.672	0.325
IMDB	128	0.696	0.046	0.680	0.317
	256	0.661	0.025	0.691	0.323

TABLE X: NetLSD embedding stability under perturbations.

Data	**D**	**Cos**	σ	**L2**	σ
	64	-0.818	0.366	52.3	30.2
MUTAG	128	-0.818	0.366	52.3	30.2
	256	0.875	0.029	216	58.7
	64	0.942	0.169	26.8	30.6
ENZ	128	0.942	0.169	26.8	30.6
	256	0.964	0.013	104	66.4
	64	0.658	0.566	58.8	61.6
IMDB	128	0.658	0.566	58.8	61.6
	256	0.962	0.027	127	83.9

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